Tucatinib, Trastuzumab and Capecitabine Therapy

INDICATIONS FOR USE:

INDICATION	ICD10	Regimen Code	HSE approved reimbursement status*
Tucatinib in combination with trastuzumab and capecitabine, for the treatment of adult patients with HER2-positive locally advanced or metastatic breast cancer who have received at least two prior anti-HER2 treatment regimens	C50	00936a	N/A

* This applies to post 2012 indications

TREATMENT:

The starting dose of the drugs detailed below may be adjusted downward by the prescribing clinician, using their independent medical judgement, to consider each patients individual clinical circumstances.

Tucatinib is administered orally twice daily, and treatment should be taken continuously. Capecitabine is administered orally twice daily on days 1-14 of each cycle. Trastuzumab is administered by intravenous (IV) infusion or by subcutaneous injection on Day 1 of each cycle.

Cycle frequency and duration:

Each cycle is 21 days. Treatment should continue until disease progression or unacceptable toxicity.

Facilities to treat anaphylaxis MUST be present when systemic anti-cancer therapy (SACT) is administered.

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Table 1. Treatment Schedule for Tucatinib, Trastuzumab (IV) and Capecitabine

Day	Drug	Dose	Route	Diluent & Rate	Cycle
1	Trastuzumab	8mg/kg	IV infusion[a] Observe post infusion[b]	250mL NaCl 0.9%[c] over 90 minutes	Cycle 1 only
1	Trastuzumab	6mg/kg	IV infusion[a] Observe post infusion[b]	250mL NaCl 0.9%[c] over 30 minutes	**Every 21 days from cycle 2 onwards**
1-21	Tucatinib[d,e,f]	300mg twice daily	PO with or without food	n/a	Continuous
1-14	Capecitabine [g, h]	1000mg/m^2 twice daily	PO with food	n/a	Every 21 days

[a]Trastuzumab can be substituted with the subcutaneous formulation where this has been approved locally. See Table 2 below.

[b]Recommended observation period: Patients should be observed for at least six hours after the start of the first infusion and for two hours after the start of the subsequent infusions for symptoms like fever and chills or other infusion-related symptoms. Any deviation should be noted in local policies.

[c]Trastuzumab is incompatible with glucose solution.

[d]Tucatinib tablets should be swallowed whole and should not be chewed, crushed, or split prior to swallowing. Tucatinib should be taken approximately 12 hours apart, at the same time every day, with or without a meal, and may be taken at the same time with capecitabine.
[e]In the case of a missed dose, the patient should take their next dose at the regularly scheduled time.
[f]Tucatinib is commonly available as 50mg and 150mg tablets

[g] The capecitabine dose to be administered should consider the tablet strengths: commonly available as 150mg and 500mg tablets
Please refer to the NCCP DOSE BANDING TABLES for dosing of capecitabine- available on the NCCP website
Tablets should be swallowed whole with plenty of water with food or within 30 minutes of eating. Tablets should not be crushed or cut. (total daily dose = 2000mg/m^2)

[h] See dose modifications section for patients with identified partial dihydropyrimidine dehydrogenase (DPD) deficiency.

Note: Administration volumes and fluids have been standardised to facilitate electronic prescribing system builds.

See alternative treatment schedule using SC trastuzumab below

Table 2. Treatment Schedule for Tucatinib, Trastuzumab (SC) and Capecitabine

Day	Drug	Dose	Route	Diluent & Rate	Cycle
1	Trastuzumab[a,b]	600mg	SC injection	Over 2-5 minutes	Every 21 days
1-21	Tucatinib[c,d,e]	300mg twice daily	PO with or without food	n/a	Continuous
1-14	Capecitabine[f,g,]	1000mg/m^2 twice daily	PO with food	n/a	Every 21 days

[a] The injection site should be alternated between the left and right thigh.
New injections should be given at least 2.5 cm from the old site and never into areas where the skin is red, bruised, tender, or hard. During the treatment course with trastuzumab subcutaneous formulation other medicinal products for subcutaneous administration should preferably be injected at different sites.

[b]Patients should be observed for 30 minutes after the first injection and for 15 minutes after subsequent injections for signs or symptoms of administration-related reactions. Any deviation should be noted in local policies.

[c]Tucatinib tablets should be swallowed whole and should not be chewed, crushed, or split prior to swallowing. Tucatinib should be taken approximately 12 hours apart, at the same time every day, with or without a meal, and may be taken at the same time with capecitabine.
[d]In the case of a missed dose, the patient should take their next dose at the regularly scheduled time.
[e]Tucatinib is commonly available as 50mg and 150mg tablets

[f] The capecitabine dose to be administered should consider the tablet strengths: commonly available as 150mg and 500mg tablets
Please refer to the NCCP DOSE BANDING TABLES for dosing of capecitabine- available on the NCCP website .
Tablets should be swallowed whole with plenty of water with food or within 30 minutes of eating. Tablets should not be crushed or cut. (total daily dose = 2000mg/m^2)

[g] See dose modifications section for patients with identified partial dihydropyrimidine dehydrogenase (DPD) deficiency.

Note: Administration volumes and fluids have been standardised to facilitate electronic prescribing system builds

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ELIGIBILITY:

- Indication as above
- HER2 overexpression or HER 2 gene amplification as determined by an accurate and validated assay. Please see *Recommendations on Reporting on HER2 Status in Breast Cancer Patients* - Available on the NCCP website
- ECOG status 0-1

CAUTIONS:

- Caution should be exercised in treating patients with increased cardiac risk, e.g. hypertension, documented coronary artery disease, CHF, LVEF of <55%, older age

EXCLUSIONS:

- Hypersensitivity to tucatinib, trastuzumab, murine proteins, capecitabine or any of the excipients
- Patients experiencing dyspnoea at rest due to complications of advanced malignancy and comorbidities may be at increased risk of a fatal infusion reaction
- Known DPD deficiency
- History of severe and unexpected reactions to fluoropyrimidine therapy
- Severe hepatic or renal impairment
- Recent or concomitant treatment with brivudine
- Pregnancy and breastfeeding

PRESCRIPTIVE AUTHORITY:

The treatment plan must be initiated by a Consultant Medical Oncologist.

TESTS:

Baseline tests:

- FBC, renal and liver profile
- Cardiac function (LVEF using ECHO, or MUGA scan) if clinically indicated
- DPD testing prior to first treatment with capecitabine using phenotype and/or genotype testing unless patient has been previously tested
    - In patients with moderate or severe renal impairment, blood uracil levels used for DPD phenotyping should be interpreted with caution, as impaired kidney function can lead to increased uracil blood levels. Consequently, there is an increased risk for incorrect diagnosis of DPD deficiency, which may result in under dosing of 5 Fluorouracil or other fluoropyrimidines, leading to reduced treatment efficacy.

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Genotype testing for DPD deficiency should be considered for patients with renal impairment

Regular tests:

- FBC and renal profile prior to each cycle
- Liver profile monitored every three weeks or as clinically indicated
- Cardiac function (ECHO or MUGA) if clinically indicated every 12 weeks. Where there are signs of cardiac impairment four to eight weekly checks may be more appropriate
- Assessment of hands and feet for palmar-plantar erythrodysaesthesia (PPE) as clinically indicated

Disease monitoring:

Disease monitoring should be in line with the patient's treatment plan and any other test/s as directed by the supervising Consultant.

DOSE MODIFICATIONS:

- Any dose modification should be discussed with a Consultant
- Please refer to Table 3 for recommended tucatinib dose reduction levels and Table 6 for recommendations on tucatinib dose modifications for adverse reactions
- **Trastuzumab IV**
    - If the patient misses a dose of trastuzumab by one week or less, then the usual maintenance dose of 6 mg/kg should be given as soon as possible. Do not wait until the next planned cycle. Subsequent maintenance doses should then be given according to the previous schedule
    - If the patient misses a dose of trastuzumab by more than one week, a re-loading dose of trastuzumab (8 mg/kg) should be given over approximately 90 minutes, at the discretion of the clinician. Subsequent trastuzumab maintenance doses (6 mg/kg) should then be given every 3 weeks from that point
- **Trastuzumab SC**
    - If the patient misses a dose of trastuzumab subcutaneous formulation, it is recommended to administer the next 600 mg dose (i.e. the missed dose) as soon as possible. The interval between consecutive Herceptin subcutaneous formulation administrations should not be less than three weeks
- **Capecitabine**
    - Consider a reduced starting dose of capecitabine in patients with identified partial DPD deficiency
        - Initial dose reduction may impact the efficacy of treatment. In the absence of serious toxicity, subsequent doses may be increased with careful monitoring

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- Toxicity due to capecitabine administration may be managed by symptomatic treatment and/or modification of the dose (treatment interruption or dose reduction)
- Once the dose has been reduced, it should not be increased at a later time
- For those toxicities considered by the treating physician to be unlikely to become serious or life-threatening, e.g. alopecia, altered taste, nail changes, treatment can be continued at the same dose without reduction or interruption
- Patients taking capecitabine should be informed of the need to interrupt treatment immediately if moderate or severe toxicity occurs
- Doses of capecitabine omitted for toxicity are not replaced

Table 3: Recommended tucatinib dose reductions for adverse reactions

Dose Level	Tucatinib Dose
Recommended starting dose	300mg twice daily
First dose reduction	250mg twice daily
Second dose reduction	200mg twice daily
Third dose reduction	150mg twice daily

Tucatinib should be permanently discontinued in patients unable to tolerate 150 mg orally twice daily.

Haematological:

- Initiation of treatment with capecitabine in patients with baseline neutrophil counts <1.5x10^9/L and/or thrombocyte counts of <100 x 10^9/L should be undertaken with caution
- If unscheduled laboratory assessments during a treatment cycle show that the neutrophil count drops below 1x10^9/L or that the platelet count drops below 75x10^9/L, treatment with capecitabine should be interrupted

Table 4: Dose modification of Capecitabine in haematological toxicity

ANC (x10^9/L)		Platelets (x10^9/L)	1st Event Dose	2nd Event Dose	3rd Event Dose	4th Event Dose
≥ 1.5	and	≥ 75	100%	100%	100%	100%
1-1.49	or	50-74.9	Delay* then 100%	Delay* then 75%	Delay* then 50%	Discontinue
0.5-0.99	or	25-49.9	Delay* then 75%	Delay* then 50%	Discontinue	Discontinue
< 0.5	or	< 25	Discontinue or delay* then 50%	Discontinue	Discontinue	Discontinue

*Delay until ANC ≥ 1.5 x 10^9/L and platelets ≥ 75 x 10^9/L

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Renal and Hepatic Impairment:

Table 5: Dose modification in renal and hepatic impairment

Drug	Renal Impairment		Hepatic Impairment	
Tucatinib*	**CrCl (mL/min)**	**Dose**		**Dose**
	≥ 30	No dose adjustment needed	Child-Pugh A/B	No dose adjustment is needed
	< 30	No need for dose adjustment is expected	Child-Pugh C	66% of the original dose
	Haemodialysis	No need for drug adjustment is expected		
Capecitabine	**CrCl (mL/min)**	**Dose**		
	51 – 80	No adjustment is needed	No dose adjustment is needed**	
	30 – 50	75% of the original dose		
	< 30	Not recommended		
	Haemodialysis	Not recommended		
Trastuzumab	**CrCl (mL/min) ***	**Dose**		
	≥ 30	No dose adjustment needed	No need for dose adjustment is expected	
	< 30	No need for dose adjustment is expected		
	Haemodialysis	No need for dose adjustment is expected		
Recommendations as per Giraud et al 2023				
***Increased creatinine without impaired renal function:** Increase in serum creatinine (30% mean increase) has been observed due to inhibition of renal tubular transport of creatinine without affecting glomerular function				
**Reference Table 7 - for dose modification of capecitabine in treatment related hepatotoxicity				

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Management of adverse events:

Tucatinib

Table 6: Dose Modification of Tucatinib for Adverse Events

Adverse reactions	Severity[1]	Recommended dose modification
Diarrhoea	Grade 1 and 2	No dose modification is required.
	Grade 3 without anti-diarrheal treatment	Initiate or intensify appropriate medical therapy. Interrupt tucatinib until recovery to ≤ Grade 1, then resume tucatinib at the same dose level.
	Grade 3 with anti-diarrheal treatment	Initiate or intensify appropriate medical therapy. Interrupt tucatinib until recovery to ≤ Grade 1, then resume tucatinib at the next lower dose level.
	Grade 4	Permanently discontinue tucatinib.
Increased ALT, AST or total bilirubin[2]	Grade 1 bilirubin (> ULN to 1.5. x ULN)	No dose modification is required.
	Grade 2 bilirubin (> 1.5 to 3 x ULN)	Interrupt tucatinib until recovery to ≤ Grade 1, then resume tucatinib at the same dose level.
	Grade 3 ALT or AST (> 5 to 20 × ULN) OR Grade 3 bilirubin (> 3 to 10 × ULN)	Interrupt tucatinib until recovery to ≤ Grade 1, then resume tucatinib at the next lower dose level.
	Grade 4 ALT or AST (> 20 × ULN) OR Grade 4 bilirubin (> 10 × ULN)	Permanently discontinue tucatinib.
	ALT or AST > 3 × ULN AND Bilirubin > 2 × ULN	Permanently discontinue tucatinib.
Other adverse reactions	Grade 1 and 2	No dose modification is required.
	Grade 3	Interrupt tucatinib until recovery to ≤ Grade 1, then resume tucatinib at the next lower dose level.
	Grade 4	Permanently discontinue tucatinib.

1. Grades based on National Cancer Institute Common Terminology Criteria for Adverse Events Version 4.03
2. Abbreviations: ULN = upper limit of normal; ALT = alanine aminotransferase; AST = aspartate aminotransferase

Capecitabine

Table 7: Dose modification of capecitabine in treatment related hepatotoxicity

Bilirubin		ALT, AST	Dose Modification
>3.0 x ULN	**or**	>2.5 X ULN	Withhold treatment until bilirubin decreases to ≤ 3.0 x ULN or ALT, AST decrease to ≤ 2.5 x ULN

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Table 8: Capecitabine dose reduction schedule (three weekly cycle) based on toxicity

Toxicity grades*	Dose changes within a treatment cycle	Dose adjustment for next cycle/dose (% of starting dose)
Grade 1	Maintain dose level	Maintain dose level
Grade 2 • 1st appearance	Interrupt until resolved to grade 0-1	100%
• 2nd appearance		75%
• 3rd appearance		50%
• 4th appearance	Discontinue permanently	
Grade 3 • 1st appearance	Interrupt until resolved to grade 0-1	75%
• 2nd appearance		50%
• 3rd appearance	Discontinue permanently	
Grade 4 • 1st appearance	Discontinue permanently or If consultant deems it to be in patient's best interest to continue, interrupt until resolved to grade 0-1	50%
• 2nd appearance	Discontinue permanently	

*Common Terminology Criteria for Adverse Events (CTCAE) version 4.0.

Table 9: Dose modification of capecitabine for diarrhoea

Grade*	Diarrhoea	Dose changes within a treatment cycle	Dose adjustment for next cycle/dose (% of starting dose)
0-1	Increase of 2 to 3 stools/day or nocturnal stools	Maintain dose level	Maintain dose level
2	Increase of 4 to 6 stools/day or nocturnal stools		
	• 1st appearance	Interrupt until resolved to grade 0-1	100%
	• 2nd appearance		75%
	• 3rd appearance		50%
	• 4th appearance	Discontinue permanently	
3	Increase of 7 to 9 stools/day or incontinence		
	• 1st appearance	Interrupt until resolved to grade 0-1	75%
	• 2nd appearance		50%
	• 3rd appearance	Discontinue permanently	
4	Increase of 10 or more stools/day or grossly bloody diarrhoea; may require parenteral support		
	• 1st appearance	Discontinue permanently	50%

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		or If consultant deems it to be in patient's best interest to continue, interrupt until resolved to grade 0-1	
	• 2nd appearance	Discontinue permanently	
Medication may be required for management of diarrhoea, e.g. loperamide (4mg at first onset followed by 2mg after each loose stool (max 16 mg /day) or see local policy			

*Common Terminology Criteria for Adverse Events (CTCAE) version 4.0.

Table 10: Dose modification of capecitabine in hand foot syndrome

Toxicity Grade		Dose Modification
Grade 1	Skin changes (e.g., numbness, dysesthesia, paresthesia, tingling, erythema) with discomfort not disrupting normal activities.	100% dose
Grade 2	Skin changes (eg, erythema, swelling) with pain affecting activities of daily living.	Withhold treatment until event resolves or decreases in intensity to grade 1.
Grade 3	Severe skin changes (eg, moist desquamation, ulceration, blistering) with pain, causing severe discomfort and inability to work or perform activities of daily living.	Withhold treatment until event resolves or decreases in intensity to grade 1. Subsequent doses of capecitabine should be decreased.

NCI CTCAE Grading

Trastuzumab

Table 11: Dose Modification of Trastuzumab for Adverse Events

Adverse reactions	Recommended dose modification
LVEF drops ≥ 10 ejection fraction points from baseline and to below 50%	Withhold treatment. Repeat LVEF after 3 weeks. No improvement or further decline consider discontinuation. Discuss with consultant and refer to cardiologist.
Symptomatic heart failure	Consider discontinuation – refer to cardiology for review. Clinical decision.
Grade 4* hypersensitivity reactions	Discontinue
Haematological	Treatment may continue during periods of reversible, chemotherapy induced myelosuppression. Monitor carefully any complications of neutropenia.

* NCI CTCAE Grading

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SUPPORTIVE CARE:

EMETOGENIC POTENTIAL:

- As outlined in NCCP Classification Document for Systemic Anti-Cancer Therapy (SACT) Induced Nausea and Vomiting Available on the NCCP website

Tucatinib: Minimal to low **(Refer to local policy)**
Capecitabine: Minimal to low **(Refer to local policy)**
Trastuzumab: Minimal **(Refer to local policy)**

For information:

Within NCIS regimens, antiemetics have been standardised by the Medical Oncologists and Haemato-oncologists and information is available in the following documents:

- NCCP Supportive Care Antiemetic Medicines for **Inclusion in NCIS** (Medical Oncology) - Available on the NCCP website
- NCCP Supportive Care Antiemetic Medicines for **Inclusion in NCIS** (Haemato-oncology) - Available on the NCCP website

PREMEDICATIONS:

- None usually required unless the patient has had a previous hypersensitivity
  - Paracetamol and antihistamine cover should be considered

OTHER SUPPORTIVE CARE:

- Medication may be required for management of diarrhoea, e.g. loperamide (4mg at first onset followed by 2mg after each loose stool (max 16 mg /day) or see local policy
- Women of childbearing potential should be advised to avoid becoming pregnant and to use effective contraception during and up to at least 1 week after treatment. Male patients with female partners of childbearing potential should also be advised to use effective contraception during and up to at least 1 week after treatment

ADVERSE EFFECTS:

- Please refer to the relevant Summary of Product Characteristics (SmPC) for details.

REGIMEN SPECIFIC COMPLICATIONS

- **Increased ALT, AST, and bilirubin** have been reported during treatment with tucatinib. ALT, AST, and total bilirubin should be monitored every three weeks or as clinically indicated. Based on the severity of the adverse reaction, treatment with tucatinib should be interrupted, then dose reduced or permanently discontinued.
- **Increased creatinine without impaired renal function:** Increase in serum creatinine (30% mean increase) has been observed due to inhibition of renal tubular transport of creatinine without affecting glomerular function. Alternative markers such as BUN, cystatin C, or calculated glomerular

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filtration rate (GFR), which are not based on creatinine, may be considered to determine whether renal function is impaired.

- **Diarrhoea,** including severe events such as dehydration, hypotension, acute kidney injury and death, has been reported during treatment with tucatinib. If diarrhoea occurs, antidiarrheals should be administered as clinically indicated. For Grade ≥ 3 diarrhoea, treatment with tucatinib 6 should be interrupted, then dose reduced or permanently discontinued. Prompt medical management should also be instituted in the event of persistence of concomitant Grade 2 diarrhoea with concomitant Grade ≥ 2 nausea and/or vomiting. Diagnostic tests should be performed as clinically indicated to exclude infectious causes of Grade 3 or 4 diarrhoea or diarrhoea of any grade with complicating features (dehydration, fever, neutropenia).
- **DPD deficiency:** DPD is an enzyme encoded by the DPYD gene which is responsible for the breakdown of fluoropyrimidines. Patients with DPD deficiency are therefore at increased risk of fluoropyrimidine-related toxicity, including for example stomatitis, diarrhoea, mucosal inflammation, neutropenia and neurotoxicity. Treatment with 5Fluorouracil, capecitabine or tegafur-containing medicinal products is contraindicated in patients with known complete DPD deficiency. Consider a reduced starting dose in patients with identified partial DPD deficiency. Initial dose reduction may impact the efficacy of treatment. In the absence of serious toxicity, subsequent doses may be increased with careful monitoring. Therapeutic drug monitoring (TDM) of 5-Fluorouracil may improve clinical outcomes in patients receiving continuous 5Fluorouracil infusions.

DRUG INTERACTIONS:

- Current SmPC and drug interaction databases should be consulted for information.

REFERENCES:

1. Murthy RK et al. Tucatinib, Trastuzumab, and capecitabine for HER2-positive metastatic breast cancer. N Engl J Med (2020); 382: 597-609. Available at https://www.nejm.org/doi/full/10.1056/NEJMoa1914609
2. Giraud E L, Lijster B D, et al. Dose recommendations for anticancer drugs in patients with renal or hepatic impairment: an update. Available at: https://pubmed.ncbi.nlm.nih.gov/37269847/
3. NCCP Classification Document for Systemic Anti-Cancer Therapy (SACT) Induced Nausea and Vomiting. V6 2025. Available at: https://www.hse.ie/eng/services/list/5/cancer/profinfo/chemoprotocols/nccp-classification-document-for-systemic-anti-cancer-therapy-sact-induced-nausea-and-vomiting.pdf
4. Tucatinib (Tukysa®). Summary of Product Characteristics. Last updated: 25/11/2024. Accessed January 2026. Available at: https://www.ema.europa.eu/en/documents/productinformation/tukysa-epar-product-information_en.pdf
5. Trastuzumab (Herceptin®) Summary of Product Characteristics EMA. Last updated 25/11/2025. Accessed January 2026. Available at: https://www.ema.europa.eu/en/documents/product-information/herceptin-epar-product-information_en.pdf

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6. Capecitabine (Xeloda®) Summary of Product Characteristics. Last updated: 02/05/2025. Accessed January 2026. Available at: https://www.ema.europa.eu/en/documents/product-information/xeloda-epar-product-information_en.pdf

Version	Date	Amendment	Approved By
1	18/02/2026		Prof Michaela Higgins

Comments and feedback welcome at oncologydrugs@cancercontrol.ie.

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